

APPENDIX B

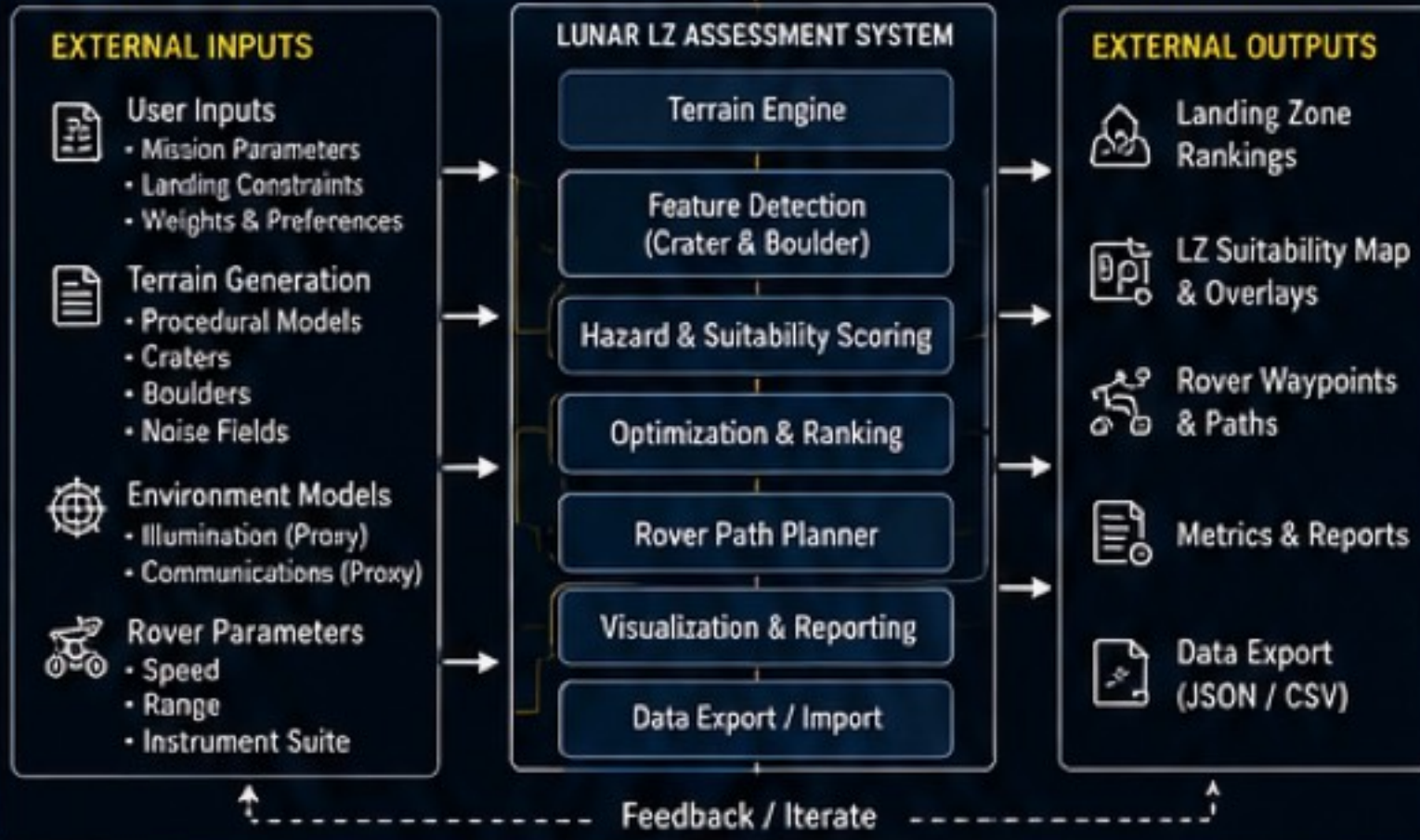
SYSTEMS ENGINEERING APPENDIX

ARCHITECTURE, REQUIREMENTS, VERIFICATION, AND RISK

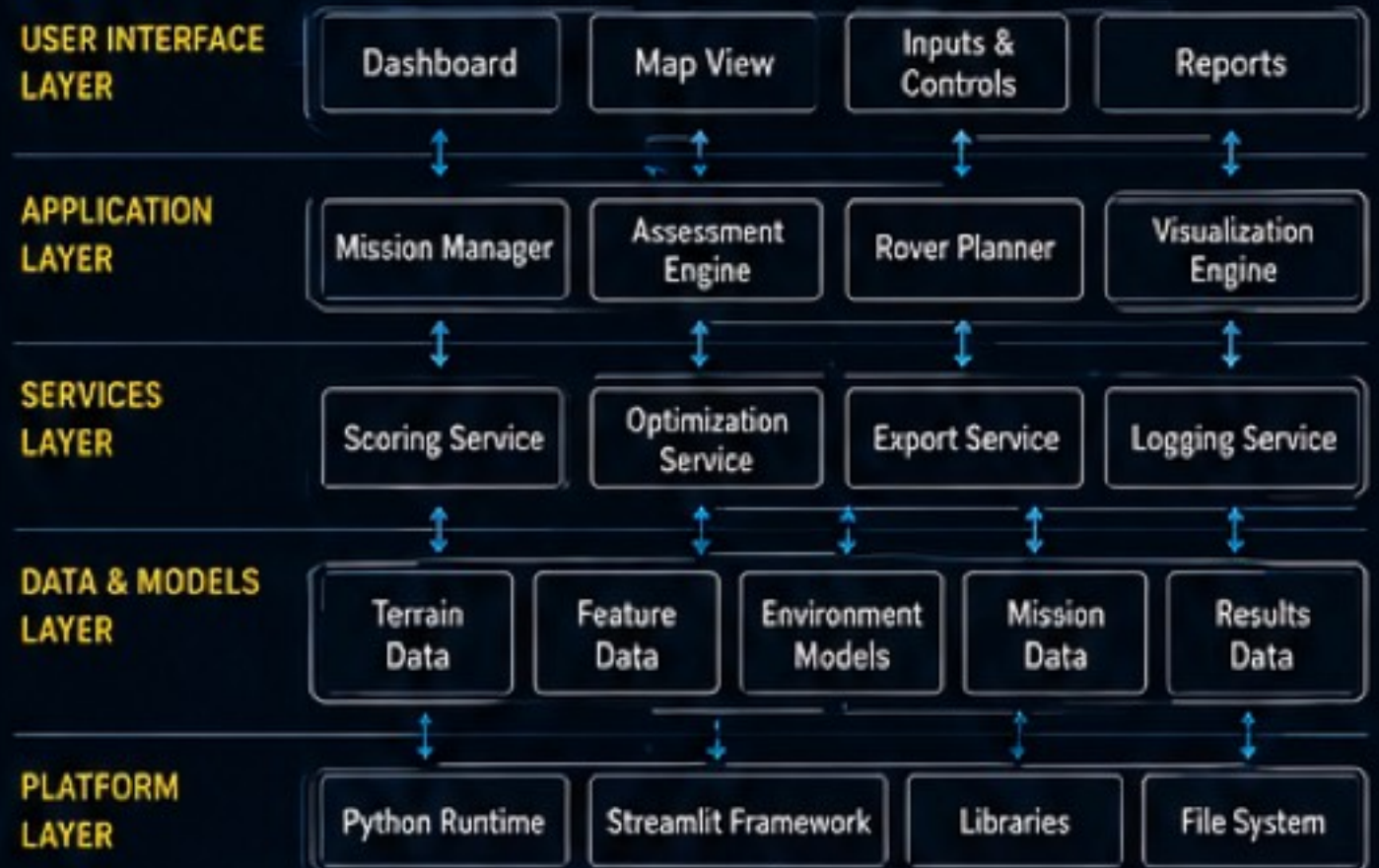
This appendix summarizes the systems engineering framework, architecture, requirements traceability, verification approach, and risk management used in the Lunar Autonomous Landing Zone Assessment application.



1 SYSTEM CONTEXT DIAGRAM



2 SYSTEM ARCHITECTURE (LOGICAL VIEW)

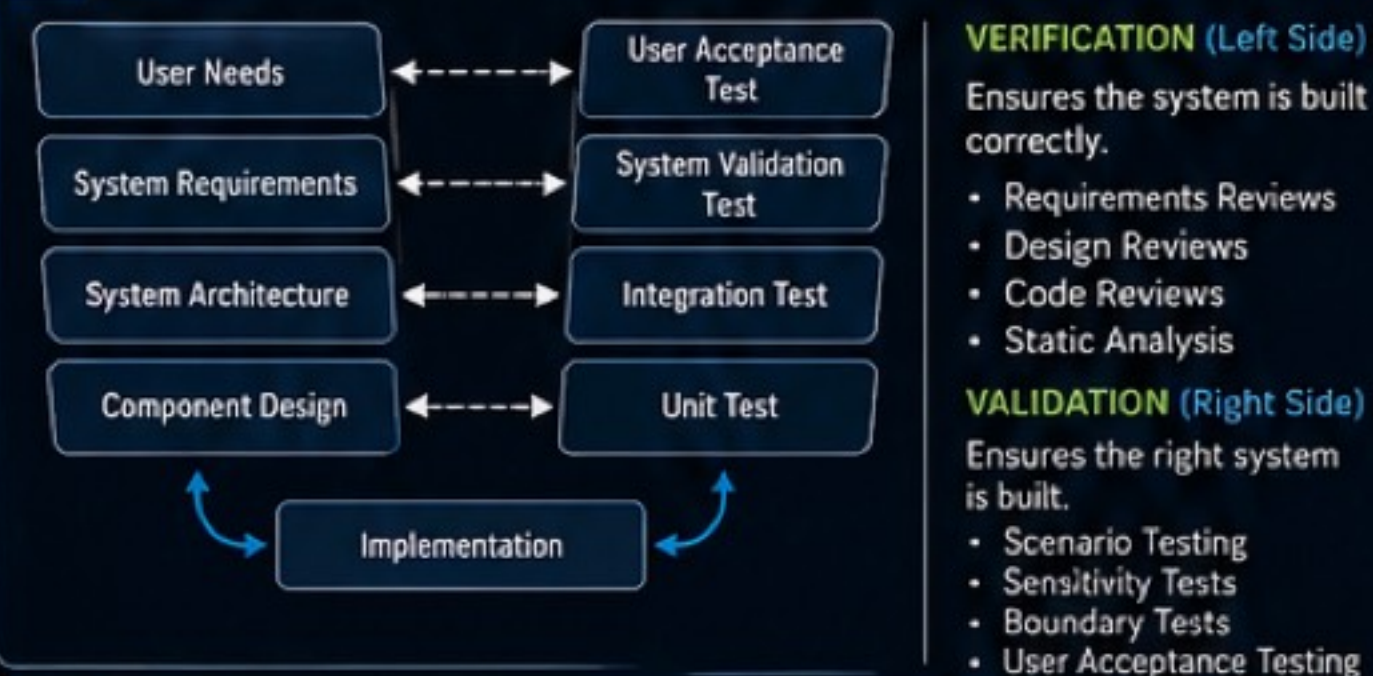


3 REQUIREMENTS SUMMARY

ID	REQUIREMENT (SYSTEM LEVEL)	TYPE	PRIORITY
REQ-1	The system shall generate a synthetic lunar terrain over a user-defined area.	FR	High
REQ-2	The system shall detect and model craters and boulders within the terrain.	FR	High
REQ-3	The system shall compute hazard and suitability scores for all candidate locations.	FR	High
REQ-4	The system shall rank candidate landing zones based on overall suitability.	FR	High
REQ-5	The system shall generate rover survey paths from a selected landing zone.	FR	Medium
REQ-6	The system shall allow users to adjust mission parameters and weighting factors.	FR	Medium
REQ-7	The system shall export results and maps in standard formats.	FR	Medium
REQ-8	The system shall provide responsive performance for areas up to 10 km x 10 km.	PR	High
REQ-9	The system shall maintain internal consistency of all calculations.	QR	High
REQ-10	The system shall log runs and support data traceability.	NFR	Medium

FR: Functional Requirement PR: Performance Requirement QR: Quality Requirement NFR: Non-Functionment

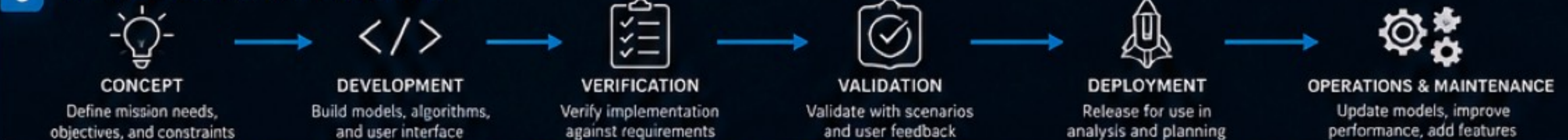
5 VERIFICATION & VALIDATION APPROACH



7 KEY PERFORMANCE PARAMETERS (EXAMPLES)

PARAMETER	DESCRIPTION	TARGET / GOAL	UNIT
Map Area	Maximum supported map size	Up to 10 km x 10 km	km
Grid Resolution	Terrain grid spacing	≤ 2 m (typical)	m
Suitability Score Range	Output landing zone score	0 to 100	score
Processing Time	End-to-end assessment time	≤ 30 s (10 km², typical)	seconds
Rover Coverage	Survey path coverage of local area	≥ 80% within 2 km radius	%
Export Formats	Supported output formats	JSON, CSV, PNG, GeoTIFF	—

9 LIFE CYCLE CONSIDERATIONS



4 REQUIREMENTS TRACEABILITY (EXAMPLE)

SYSTEM NEED	REQUIREMENT	DESIGN ELEMENT	VERIFICATION METHOD
Safe Landing Site Identification	REQ-1, REQ-2, REQ-3, REQ-4	Terrain Engine, Hazard & Suitability Scoring, Optimizer	Model Verification, Unit Tests, Scenario Tests
Mission Flexibility	REQ-6	Inputs & Controls, Mission Manager	Functional Test, User Acceptance Test
Rover Reconnaissance	REQ-5	Rover Path Planner	Path Validation, Coverage Analysis
Data Management	REQ-7, REQ-10	Export Service, Logging Service	Export Validation, Traceability Check
Performance & Reliability	REQ-8, REQ-9	Platform, Services, Codebase	Performance Test, Consistency Checks

6 RISK MANAGEMENT SUMMARY

RISK ID	RISK DESCRIPTION	CAUSE	MITIGATION	SEVERITY
R-1	Terrain model not representative	Procedural simplifications	Calibrate parameters, compare with real data	High
R-2	Missed or inaccurate hazard detection	Feature model limits, resolution	Increase resolution, improve detection	High
R-3	Over-reliance on proxy environment models	Simplified solar/comm calculations	Use as screening, not certification	Medium
R-4	Weighting factors chosen poorly	User bias / uncertainty	Provide guidance, sensitivity analysis	Medium
R-5	Performance degradation on large maps	High resolution, complex scoring	Optimize code, use efficient structures	Low
R-6	Data export errors	Format / I/O issues	Validation checks, automated tests	Low

High: Requires immediate attention Medium: Monitor and mitigate Low: Acceptable with controls

8 ASSUMPTIONS & LIMITATIONS

ASSUMPTIONS

- Lunar surface is airless and unweathered.
- Procedural terrain approximates regolith.
- Illumination model is a relative proxy.
- Communication model is geometry-based.
- Rover operates on nonimal terrain.
- User provides accurate mission inputs.

LIMITATIONS

- Not a flight-certified system.
- Does not model lander plume interaction.
- Does not model regolith mechanics or bearing strength.
- Does not model thermal extremes in detail.
- Not a substitute for high-fidelity mission analysis.